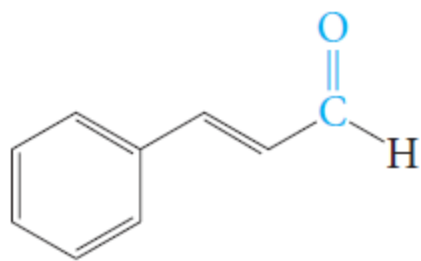
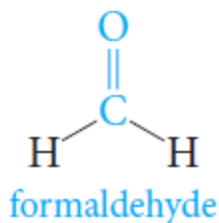


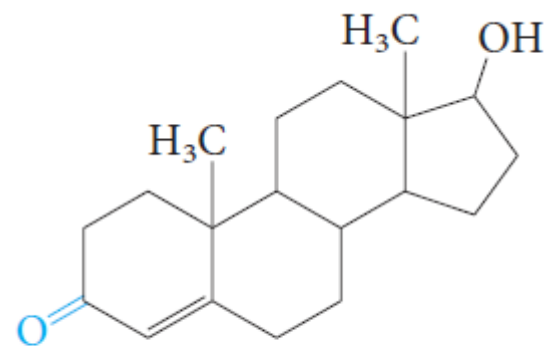
Aldehydes and Ketones



cinnamaldehyde



formaldehyde



testosterone

Chemistry Dept., JUST

Carbonyl Compounds

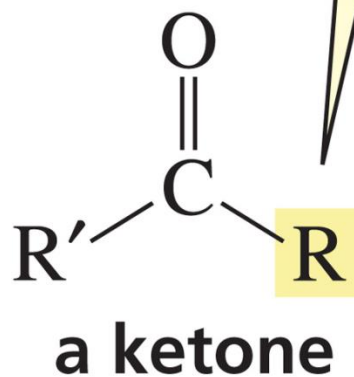
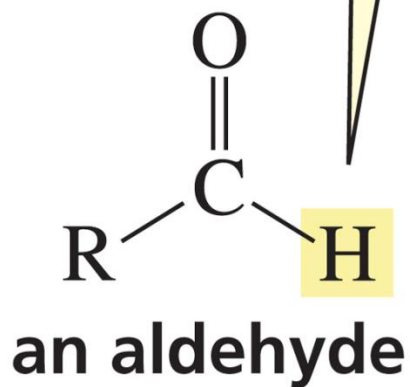


All carbonyl compounds contain the acyl group

The group X is strongly affect the chemistry and reactivity of the carbonyl compounds

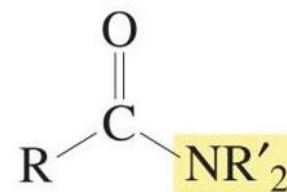
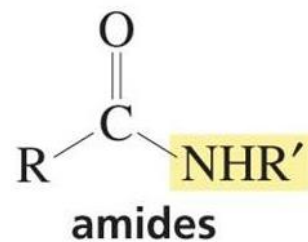
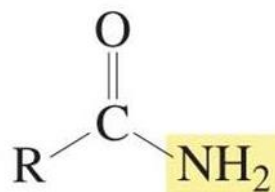
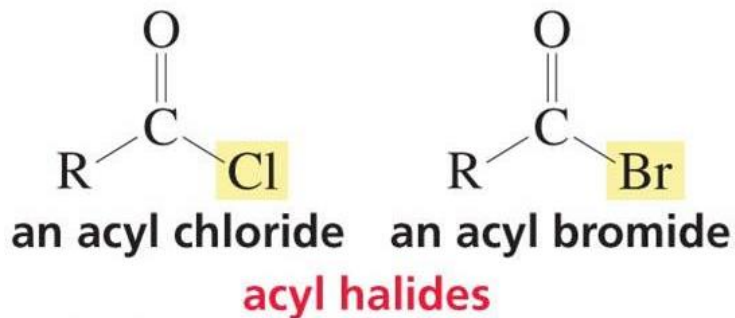
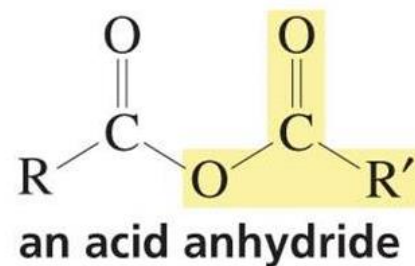
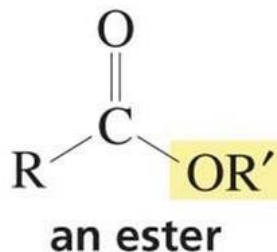
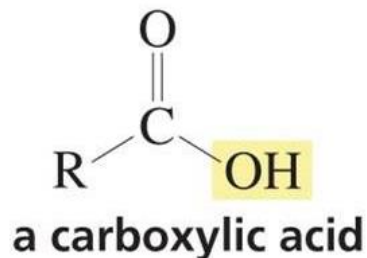
Carbonyl Compounds

cannot be replaced by a nucleophile

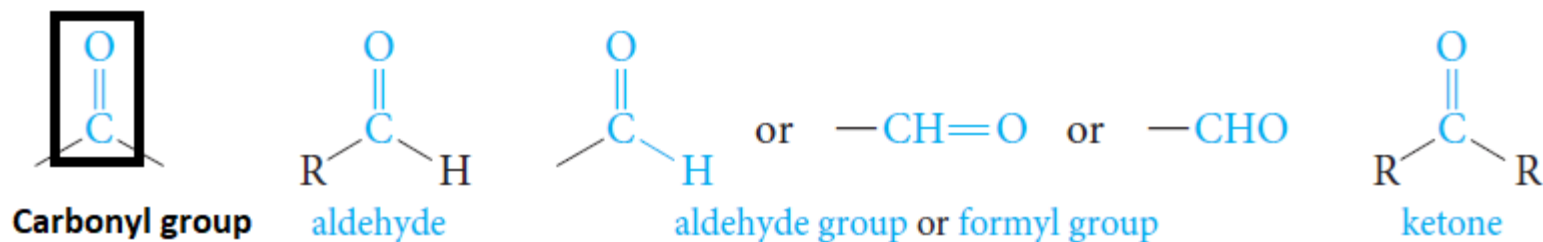


Carbonyl Compounds

carbonyl compounds with groups that can be replaced by a nucleophile

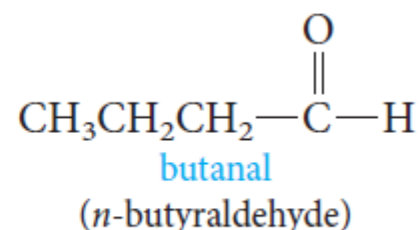
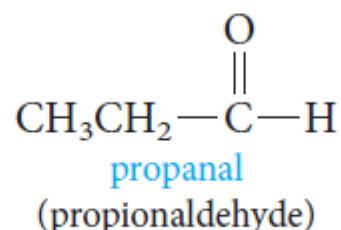
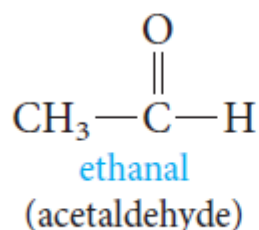
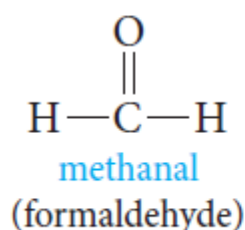


Aldehydes and ketones are characterized by the presence of the **carbonyl group**, perhaps the most important functional group in organic chemistry. Aldehydes have at least one hydrogen atom attached to the carbonyl carbon atom. The remaining group may be another hydrogen atom or any aliphatic or aromatic organic group. The CHO group characteristic of aldehydes is often called a **formyl group**. In **ketones**, the carbonyl carbon atom is connected to two other carbon atoms

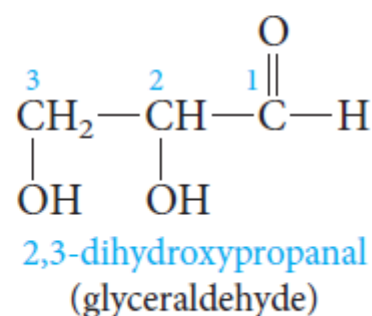
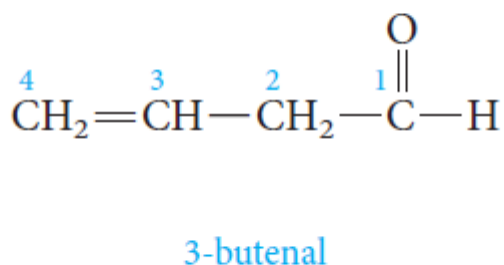
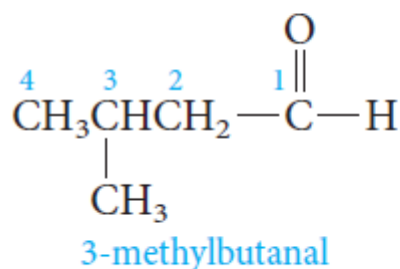


9.1 Nomenclature of Aldehydes and Ketones

In the IUPAC system, the characteristic ending for aldehydes is *-al* (from the first syllable of *aldehyde*). The following examples illustrate the system:



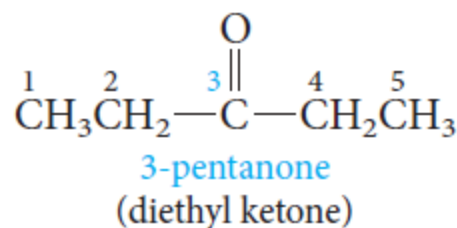
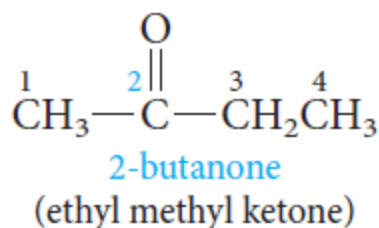
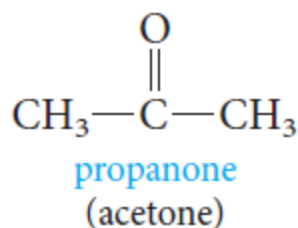
For substituted aldehydes, we number the chain starting with the aldehyde carbon, as the following examples illustrate:

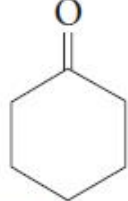


9.1 Nomenclature of Aldehydes and Ketones

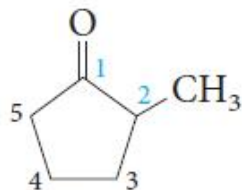
In the IUPAC system, the ending for ketones is *-one* (from the last syllable of *ketone*). The chain is numbered so that the carbonyl carbon has the lowest possible number. Common names of ketones are formed by adding the word *ketone* to the names of

the alkyl or aryl groups attached to the carbonyl carbon. In still other cases, traditional names are used. The following examples illustrate these methods:

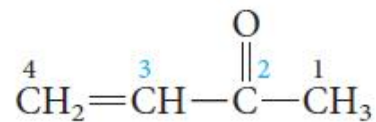




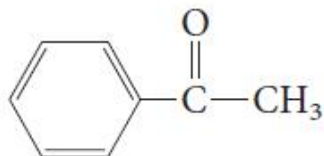
cyclohexanone



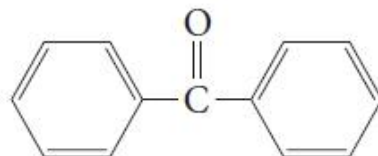
2-methylcyclopentanone



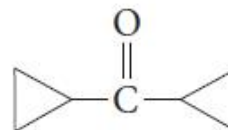
3-buten-2-one
(methyl vinyl ketone)



acetophenone
(methyl phenyl ketone)



benzophenone
(diphenyl ketone)



dicyclopropyl ketone

PROBLEM 9.1 Using the examples as a guide, write a structure for

- | | |
|----------------------------|---------------------------------|
| a. hexanal | b. <i>o</i> -chlorobenzaldehyde |
| c. 4-heptanone | d. isopropyl methyl ketone |
| e. cyclobutanecarbaldehyde | f. 3-octyn-2-one |

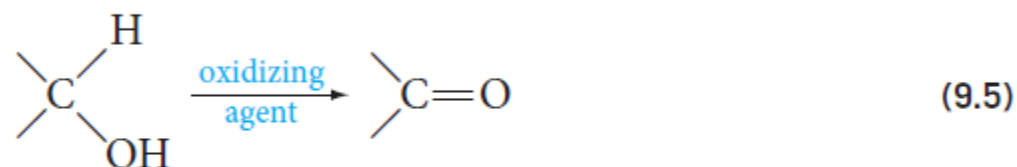
PROBLEM 9.2 Using the examples as a guide, write a correct name for

- | | |
|--|--|
| a. $(\text{CH}_3)_2\text{C}(\text{Br})\text{CH}_2\text{CH}=\text{O}$ | b. $\text{CH}_3\text{CH}=\text{CHCH}=\text{O}$ |
|--|--|



9.3 Synthesis of Aldehydes and Ketones

We have already seen, in previous chapters, several ways to prepare aldehydes and ketones. One of the most useful is the oxidation of alcohols.



Oxidation of a primary alcohol gives an aldehyde, and oxidation of a secondary alcohol gives a ketone. Chromium reagents, such as pyridinium chlorochromate (PCC), are commonly used in the laboratory for this purpose (review Sec. 7.12).

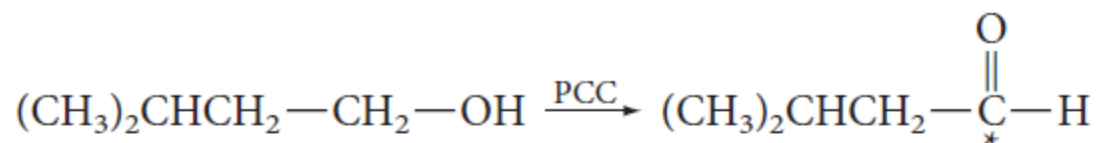
PROBLEM 9.3 Give the product expected from treatment of

- cyclohexanol with Jones' reagent (see Sec. 7.12)
- 3-methylcyclopentanol with pyridinium chlorochromate
- 5-methyl-1-heptanol with Jones' reagent
- 5-methyl-1-heptanol with pyridinium chlorochromate

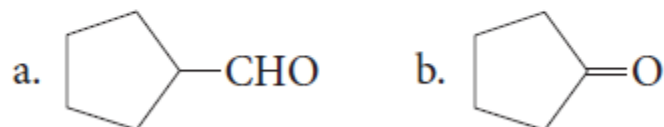
EXAMPLE 9.1

Write an equation for the oxidation of an appropriate alcohol to $(\text{CH}_3)_2\text{CHCH}_2\text{CHO}$ (3-methylbutanal).

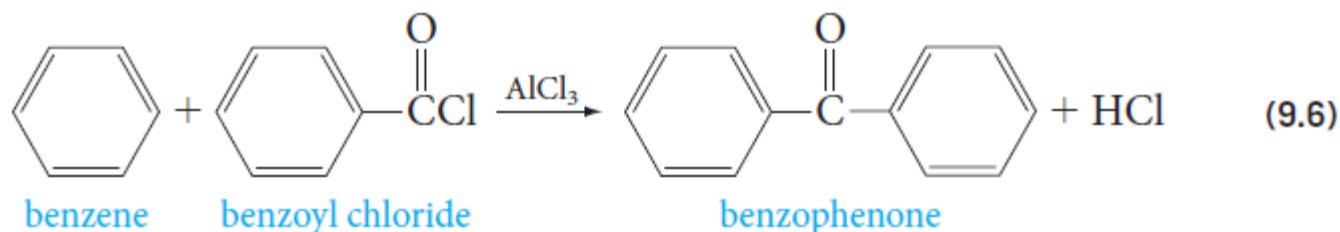
Solution Aldehydes are prepared by oxidation of 1° alcohols (RCH_2OH) with PCC. First, find the carbonyl carbon in 3-methylbutanal (marked with an asterisk). Convert this carbon to a primary alcohol. A proper equation is:



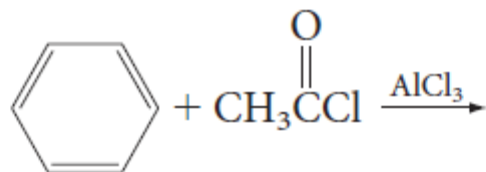
PROBLEM 9.4 Write an equation for the oxidation of an appropriate alcohol to



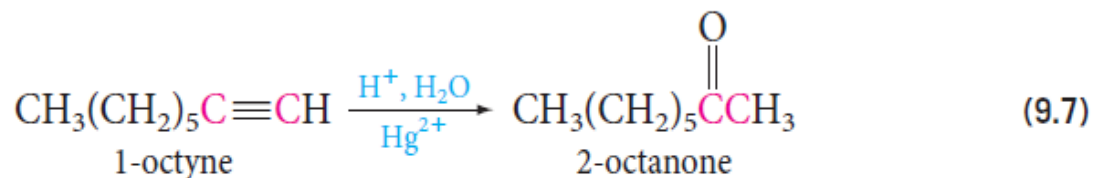
Aromatic ketones can be prepared by Friedel–Crafts acylation of an aromatic ring (review eq. 4.12 and Sec. 4.9.d). For example,



PROBLEM 9.5 Complete the following equation and name the product:



Methyl ketones can be prepared by hydration of terminal alkynes, catalyzed by acid and mercuric ion (review eq. 3.52). For example,



PROBLEM 9.6 What alkyne would be useful for the synthesis of 2-heptanone (oil of cloves)?



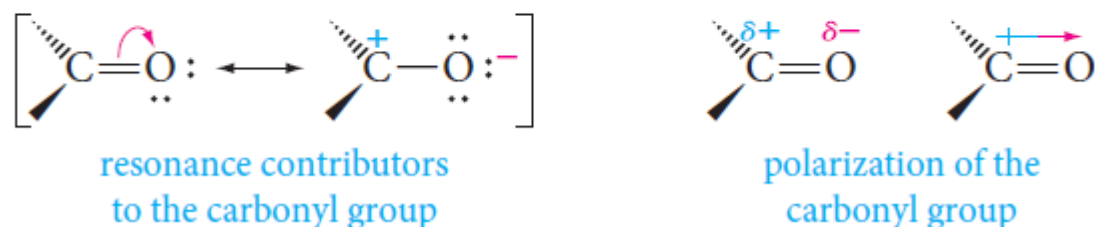
Cloves, source of 2-heptanone.

9.5 The Carbonyl Group

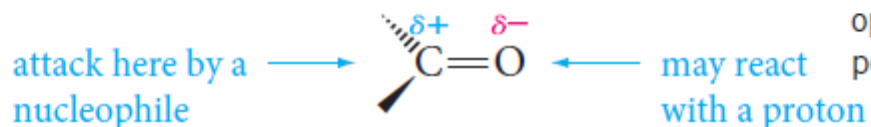
To best understand the reactions of aldehydes, ketones, and other carbonyl compounds, we must first appreciate the structure and properties of the carbonyl group.

The carbon–oxygen double bond consists of a sigma bond and a pi bond (Figure 9.2). The carbon atom is sp^2 -hybridized. *The three atoms attached to the carbonyl carbon lie in a plane with bond angles of 120° .* The pi bond is formed by overlap of a p orbital on carbon with an oxygen p orbital. There are also two unshared electron pairs on the oxygen atom. The C=O bond distance is 1.24 Å, shorter than the C—O distance in alcohols and ethers (1.43 Å).

Oxygen is much more electronegative than carbon. Therefore, the electrons in the C=O bond are attracted to the oxygen, producing a highly polarized bond. This effect is especially pronounced for the pi electrons and can be expressed in the following ways:



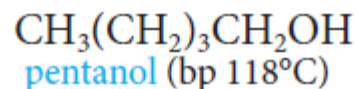
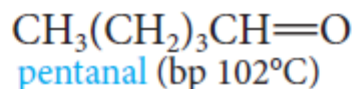
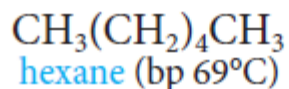
As a consequence of this polarization, *most carbonyl reactions involve nucleophilic attack at the carbonyl carbon*, often accompanied by addition of a proton to the oxygen, consistent with the electrostatic potential map (Figure 9.2c) showing that the oxygen is electron rich (red).



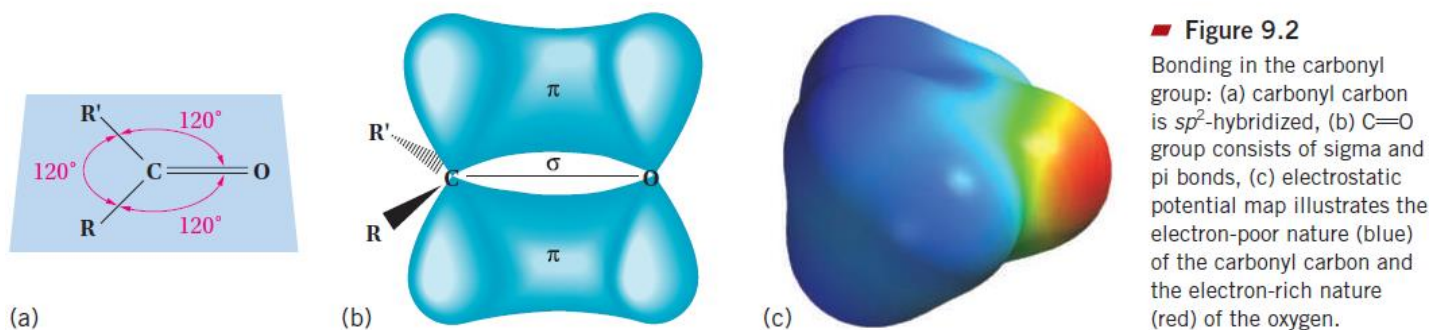
Dipole–dipole interactions are opposite pole attractions between polar molecules.

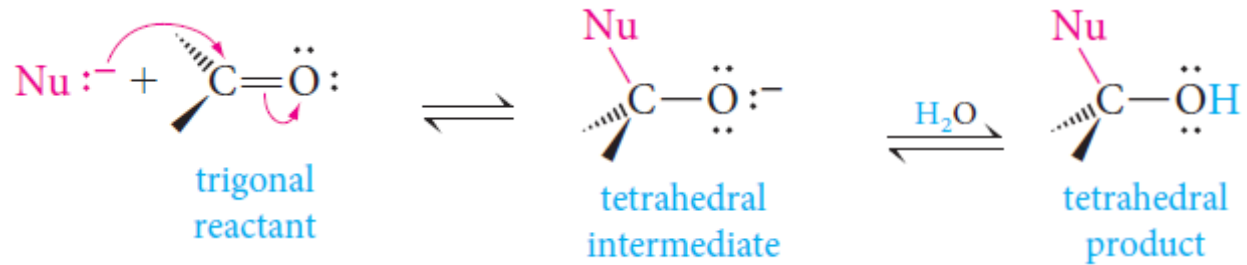
C=O bonds are quite different, then, from C=C bonds, which are not polarized and where attack at carbon is usually by an electrophile (Sec. 3.9).

In addition to its effect on reactivity, polarization of the C=O bond influences the physical properties of carbonyl compounds. For example, carbonyl compounds boil at higher temperatures than hydrocarbons, but at lower temperatures than alcohols of comparable molecular weight.



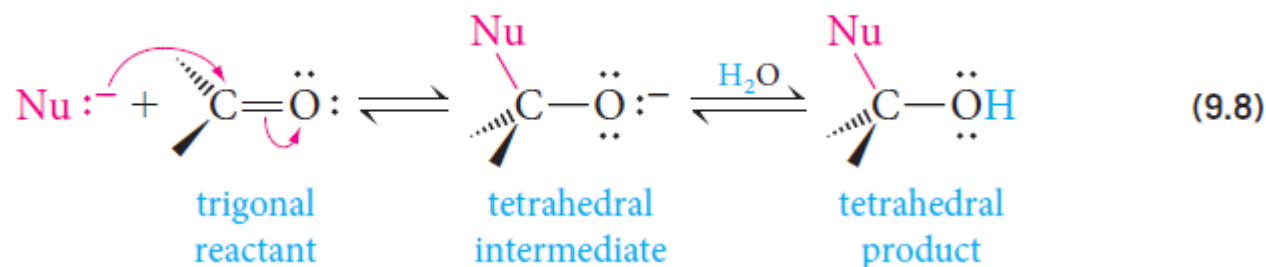
Why is this so? Unlike hydrocarbon molecules, which can be temporarily polarized, molecules of carbonyl compounds have *permanently polar C=O bonds* and thus have a stronger tendency to associate. *The positive part of one molecule is attracted to the negative part of another molecule.* These intermolecular forces of attraction, called **dipole–dipole interactions**, are generally stronger than van der Waals attractions (Sec. 2.7) but not as strong as hydrogen bonds (Sec. 7.4). Carbonyl compounds such





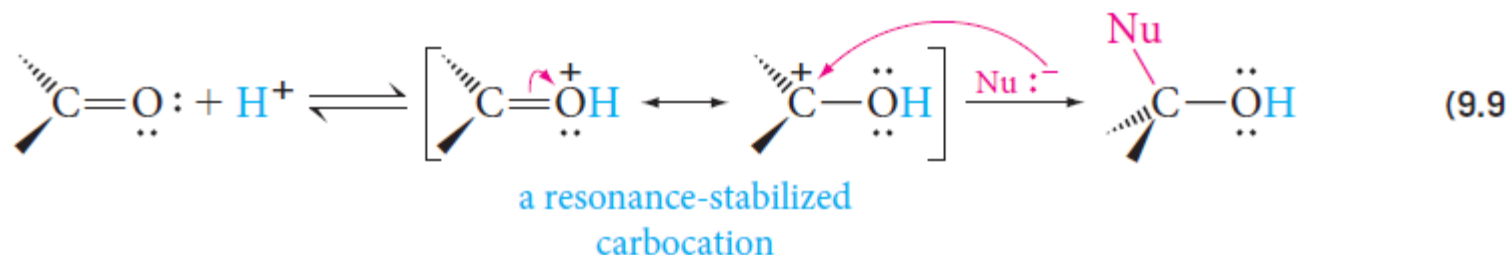
Nucleophilic Addition to Carbonyl Groups: An Overview

Nucleophiles attack the carbon atom of a carbon–oxygen double bond because that carbon has a partial positive charge. The pi electrons of the C=O bond move to the oxygen atom, which, because of its electronegativity, can easily accommodate the negative charge that it acquires. When these reactions are carried out in a hydroxylic solvent such as alcohol or water, the reaction is usually completed by addition of a proton to the negative oxygen. The overall reaction involves addition of a nucleophile and a proton across the pi bond of the carbonyl group.



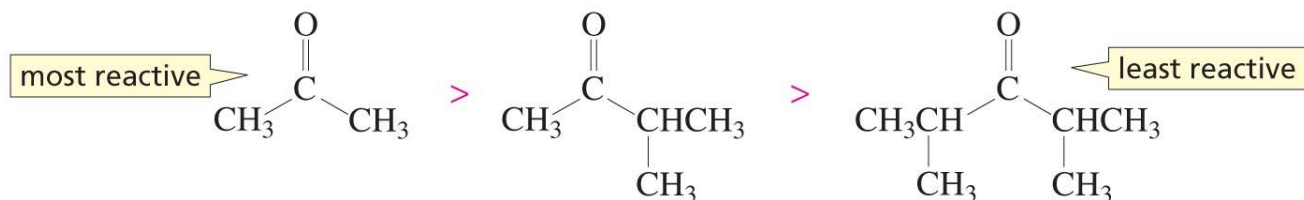
The carbonyl carbon, which is trigonal and sp^2 -hybridized in the starting aldehyde or ketone, becomes tetrahedral and sp^3 -hybridized in the reaction product.

Because of the unshared electron pairs on the oxygen atom (Figure 9.2c), carbonyl compounds are weak Lewis bases and can be protonated. *Acids can catalyze the addition of weak nucleophiles to carbonyl compounds* by protonating the carbonyl oxygen atom.



In general, *ketones are somewhat less reactive than aldehydes toward nucleophiles*. There are two main reasons for this reactivity difference. The first reason is *steric*. The carbonyl carbon atom is more crowded in ketones (two organic groups) than in aldehydes (one organic group and one hydrogen atom). In nucleophilic addition, we bring these attached groups closer together because the hybridization changes from sp^2 to sp^3 and the bond angles decrease from 120° to 109.5° (eq. 9.8). Less strain is involved in additions to aldehydes than in additions to ketones because one of the groups (H) is small.

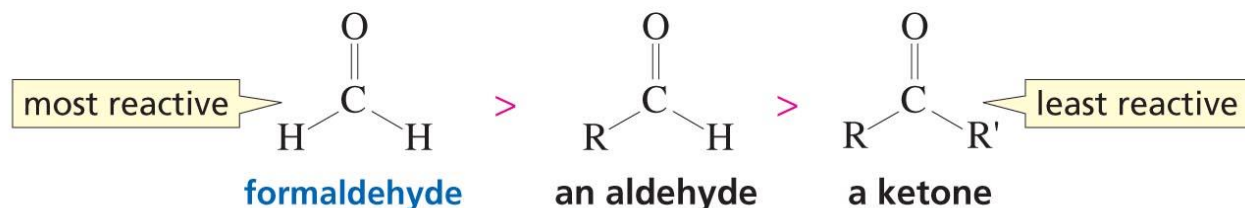
relative reactivities



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. The second reason is *electronic*. As we have already seen in connection with carbocation stability, alkyl groups are usually electron-donating compared to hydrogen (Sec. 3.10). Therefore, they tend to neutralize the partial positive charge on the carbonyl carbon, decreasing its reactivity toward nucleophiles. Ketones have two such alkyl groups; aldehydes have only one. If, however, the attached groups are strongly electron-withdrawing (contain halogens, for example), they can have the opposite effect and increase carbonyl reactivity toward nucleophiles.

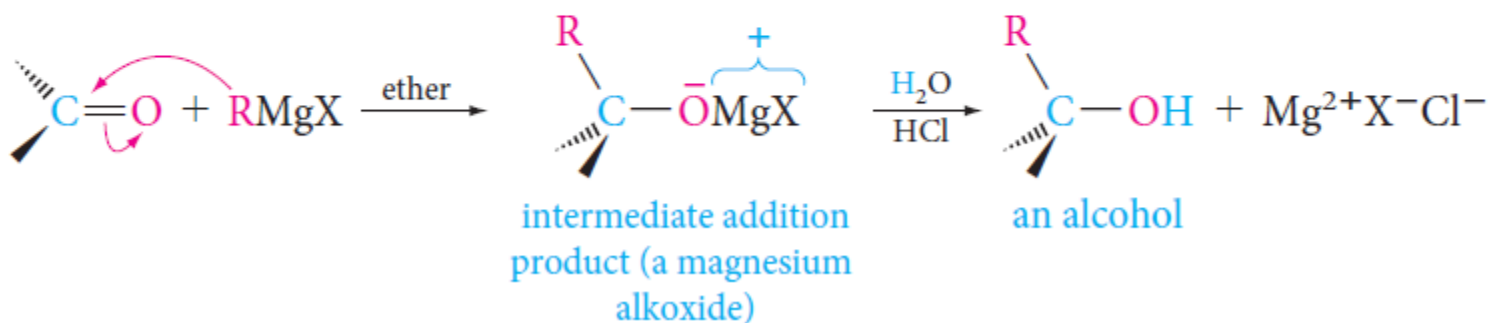
relative reactivities



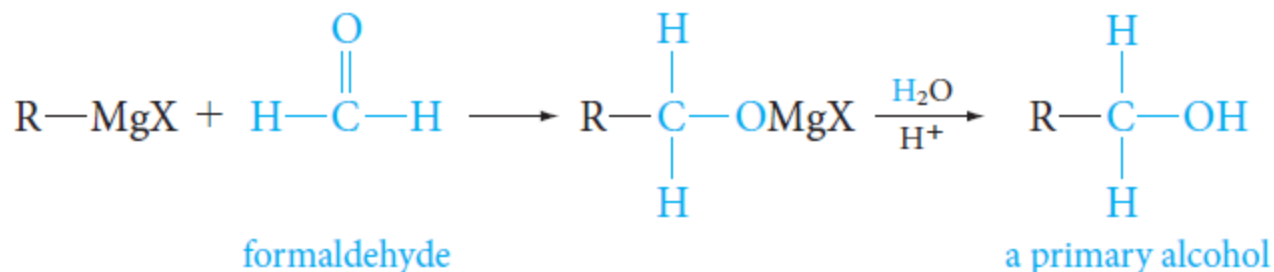
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Addition of Grignard Reagents and Acetylides

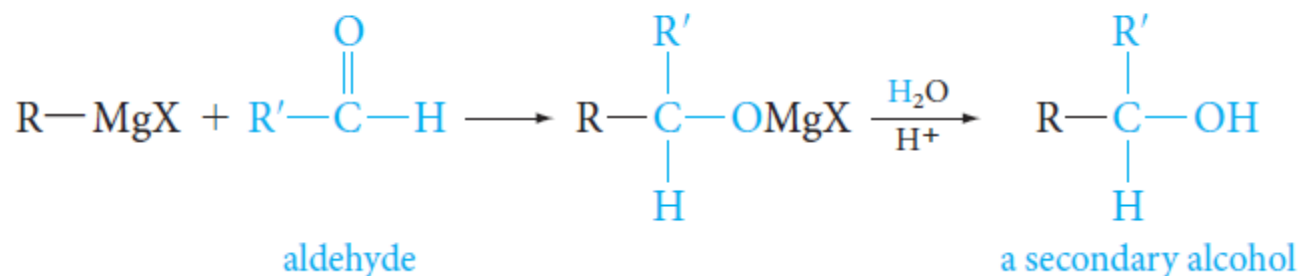
Grignard reagents act as carbon nucleophiles toward carbonyl compounds. The R group of the Grignard reagent adds irreversibly to the carbonyl carbon, forming a new carbon-carbon bond. In terms of acid-base reactions, the addition is favorable because the product (an alkoxide) is a much weaker base than the starting carbanion (Grignard reagent). The alkoxide can be protonated to give an alcohol.



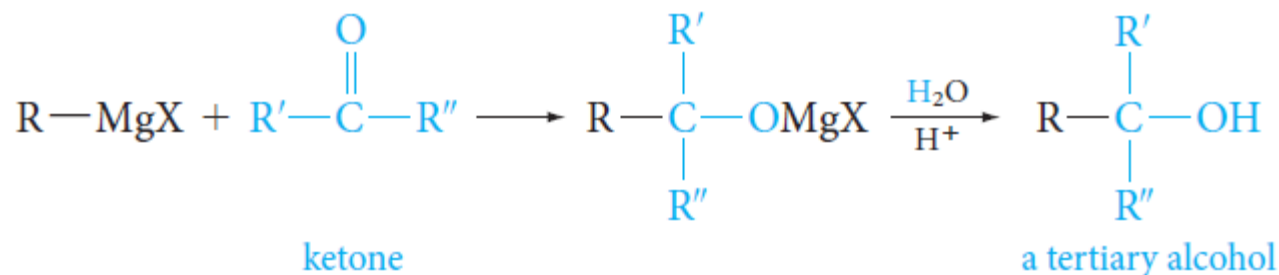
Formaldehyde gives primary alcohols.



Other aldehydes give secondary alcohols.



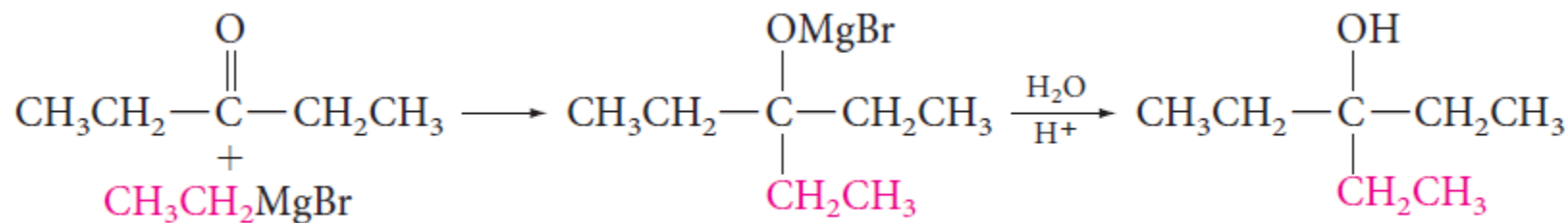
Ketones give tertiary alcohols.



EXAMPLE 9.4

What is the product expected from the reaction between ethylmagnesium bromide and 3-pentanone followed by hydrolysis?

Solution 3-Pentanone is a ketone. Following eq. 9.24 as an example, the product is 3-ethyl-3-pentanol.

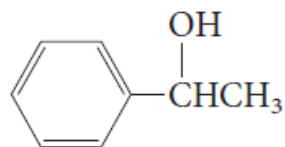


PROBLEM 9.14 Provide the products expected from the reaction of

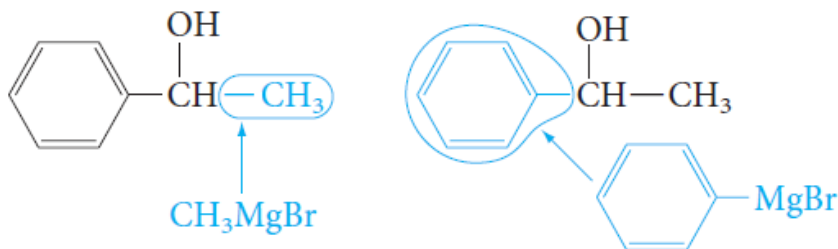
- formaldehyde with benzylmagnesium bromide followed by hydrolysis
- pentanal with phenylmagnesium bromide followed by hydrolysis
- benzaldehyde with ethylmagnesium bromide followed by hydrolysis
- cyclohexanone with methylmagnesium bromide followed by hydrolysis

EXAMPLE 9.5

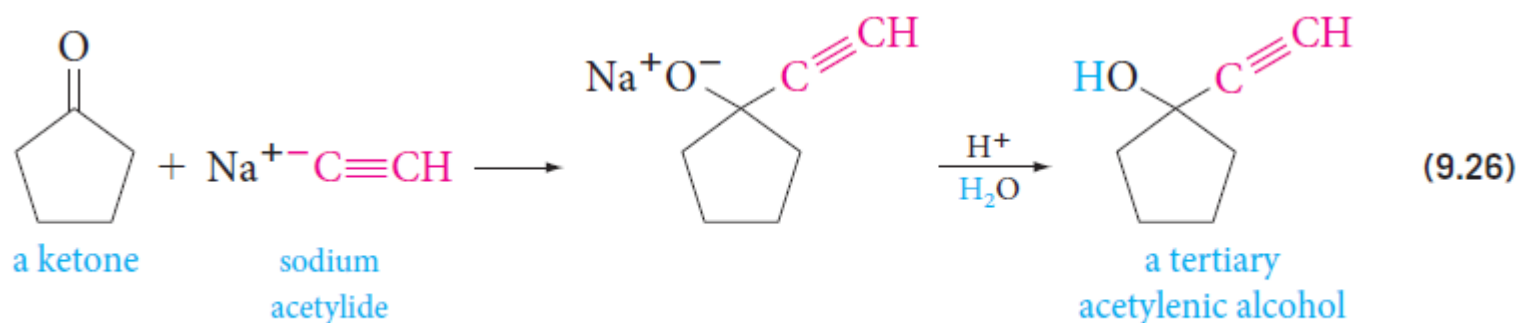
Show how the following alcohol can be synthesized from a Grignard reagent and a carbonyl compound:



Solution The alcohol is secondary, so the carbonyl compound must be an aldehyde. We can use either a methyl or a phenyl Grignard reagent.



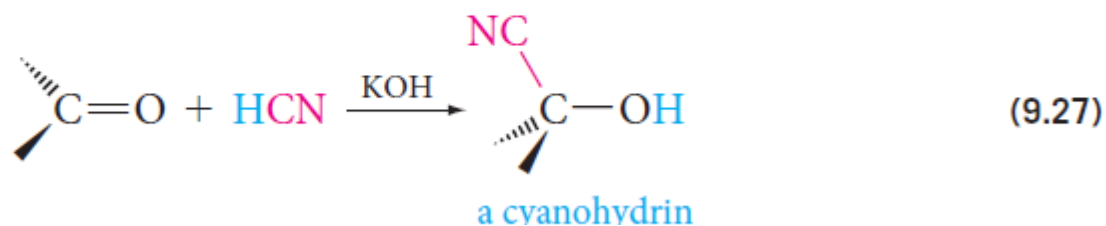
Other organometallic reagents, such as organolithium compounds and acetylides, react with carbonyl compounds in a similar fashion to Grignard reagents. For example,



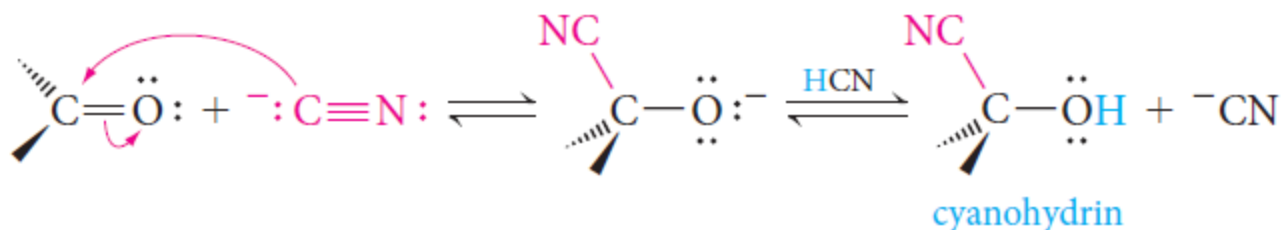
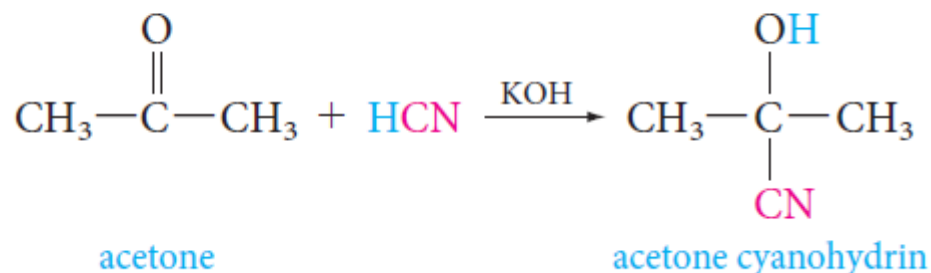
9.10

Addition of Hydrogen Cyanide; Cyanohydrins

Hydrogen cyanide adds reversibly to the carbonyl group of aldehydes and ketones to form **cyanohydrins**, compounds with a hydroxyl and a cyano group attached to the same carbon. A basic catalyst is required.



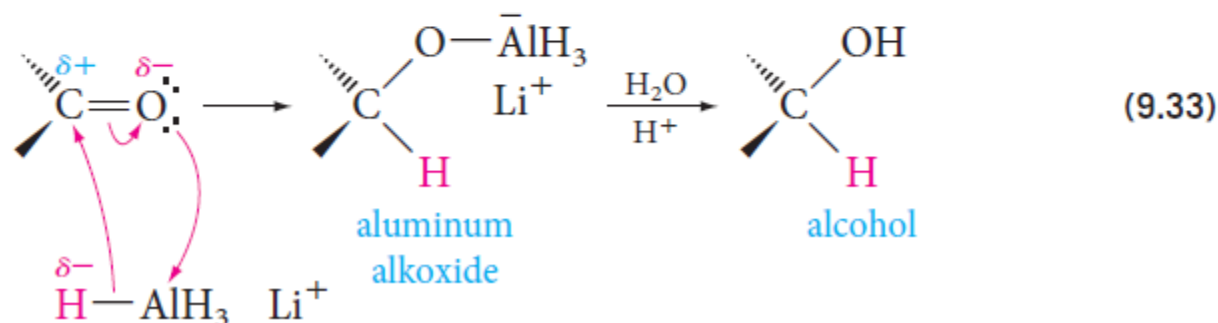
Acetone, for example, reacts as follows:



9.12 Reduction of Carbonyl Compounds

Aldehydes and ketones are easily reduced to primary and secondary alcohols, respectively. Reduction can be accomplished in many ways, most commonly by metal hydrides.

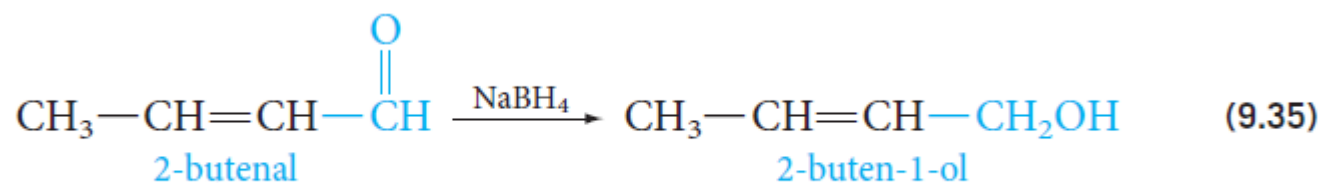
The most common metal hydrides used to reduce carbonyl compounds are lithium aluminum hydride (LiAlH_4) and sodium borohydride (NaBH_4). The metal-hydride bond is polarized, with the metal positive and the hydrogen negative. Therefore, the reaction involves irreversible nucleophilic attack of the hydride (H^-) at the carbonyl carbon:



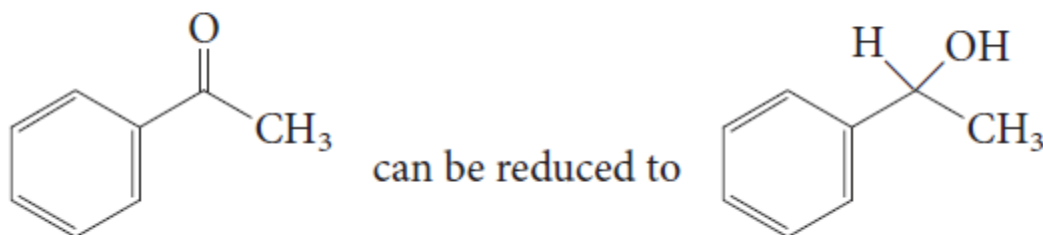
The initial product is an aluminum alkoxide, which is subsequently hydrolyzed by water and acid to give the alcohol. The net result is addition of hydrogen across the carbon-oxygen double bond. A specific example is



Because a carbon–carbon double bond is not readily attacked by nucleophiles, metal hydrides can be used to reduce a carbon–oxygen double bond to the corresponding alcohol without reducing a carbon–carbon double bond present in the same compound.

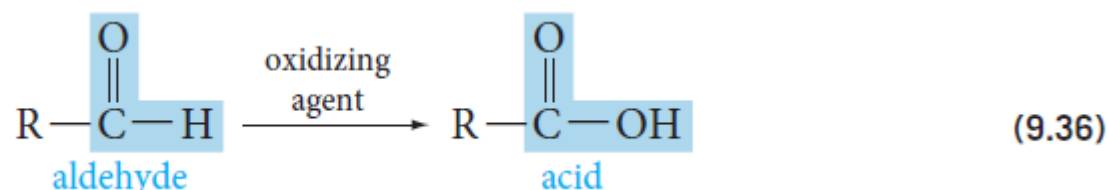


PROBLEM 9.21 Show how

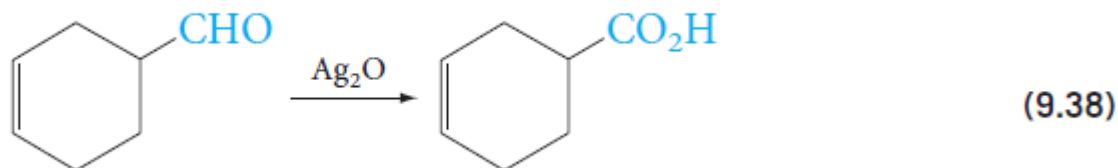
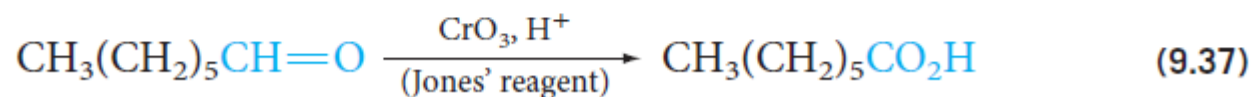


9.13 Oxidation of Carbonyl Compounds

Aldehydes are more easily oxidized than ketones. Oxidation of an aldehyde gives a carboxylic acid with the same number of carbon atoms.

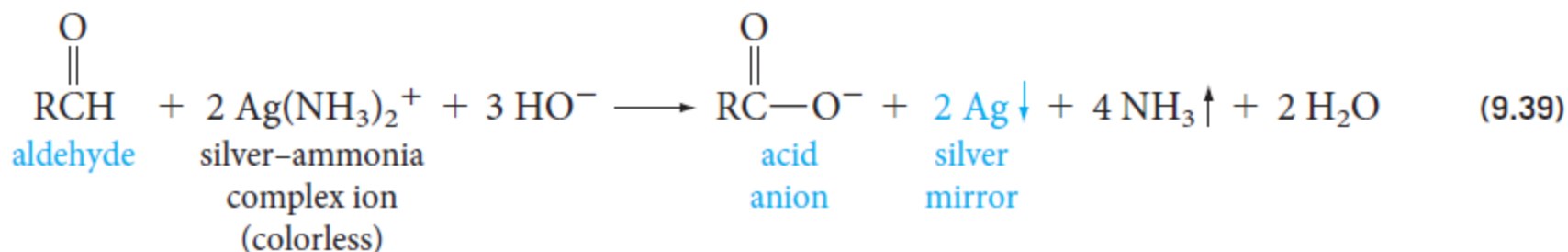


Because the reaction occurs easily, many oxidizing agents, such as KMnO_4 , CrO_3 , Ag_2O , and peracids (see eq. 8.18), will work. Specific examples are



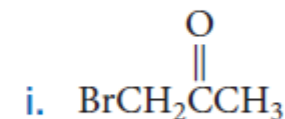
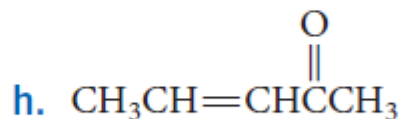
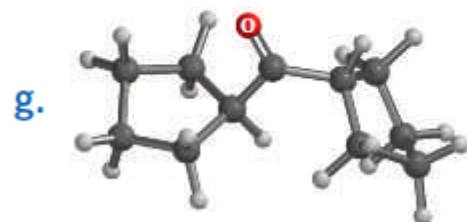
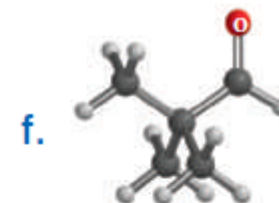
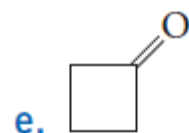
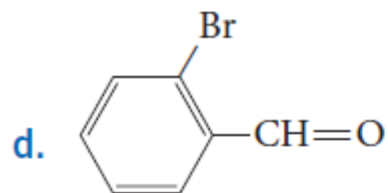
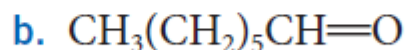
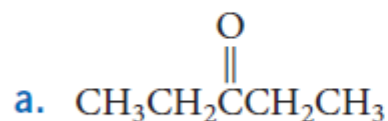
Silver ion as an oxidant is expensive but has the virtue that it selectively oxidizes aldehydes to carboxylic acids in the presence of alkenes (eq. 9.38).

A laboratory test that distinguishes aldehydes from ketones takes advantage of their different ease of oxidation. In the Tollens' silver mirror test, the silver–ammonia complex ion is reduced by aldehydes (but not by ketones) to metallic silver.* The equation for the reaction may be written as follows:



Nomenclature, Structure, and Properties of Aldehydes and Ketones

9.30 Name each of the following compounds.



9.31 Write a structural formula for each of the following:

a. 3-heptanone

b. 3-isopropylheptanal

c. *p*-bromobenzaldehyde

d. (*S*)-2-methylcyclohexanone

e. 3-hexenal

f. benzyl *p*-methylphenyl ketone

g. 2,2-dichlorohexanal

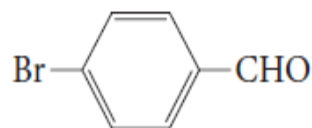
h. 4-phenyl-2-butanone

i. cyclobutanone

j. *p*-tolualdehyde

Reactions of Aldehydes and Ketones

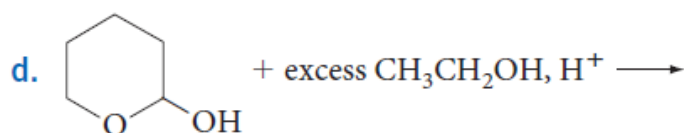
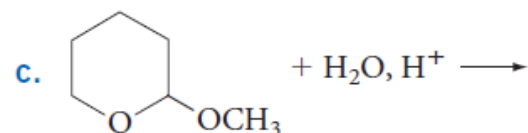
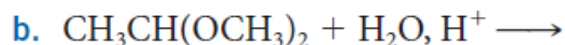
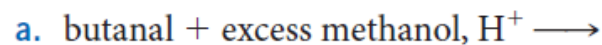
9.38 Write an equation for the reaction, if any, of *p*-bromobenzaldehyde with each of the following reagents, and name the organic product.



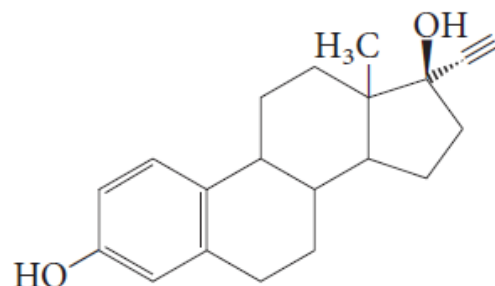
p-bromobenzaldehyde

- | | |
|---|---|
| a. methylmagnesium bromide, then H_3O^+ | b. methylamine (CH_3NH_2) |
| c. ethylene glycol, H^+ | d. phenylhydrazine |
| e. HCN (catalytic NaOH) | f. Tollens' reagent |

9.41 Complete each of the following equations:



9.42 Estrone (page 253) can be readily converted to ethynylestradiol:



Ethynylestradiol was the first synthetic estrogen to be used in formulations of oral contraceptive pills. Suggest a synthesis of ethynylestradiol from estrone.

Reactions with Grignard Reagents and Other Nucleophiles

9.44 Write an equation for the reaction of each of the following with ethylmagnesium bromide, followed by hydrolysis with aqueous acid:

- a. formaldehyde
- b. hexanal
- c. acetophenone
- d. ethylene oxide
- e. cyclohexanone

9.45 Using a Grignard reagent and the appropriate aldehyde or ketone, show how each of the following can be prepared:

- a. methylcyclohexanol
- b. 3-octanol
- c. 2-methyl-2-pentanol
- d. 1-cyclopentylcyclopentanol
- e. 1-phenyl-1-propanol
- f. 3-butene-2-ol

Oxidations and Reductions

9.48 Give the structure of each product.

